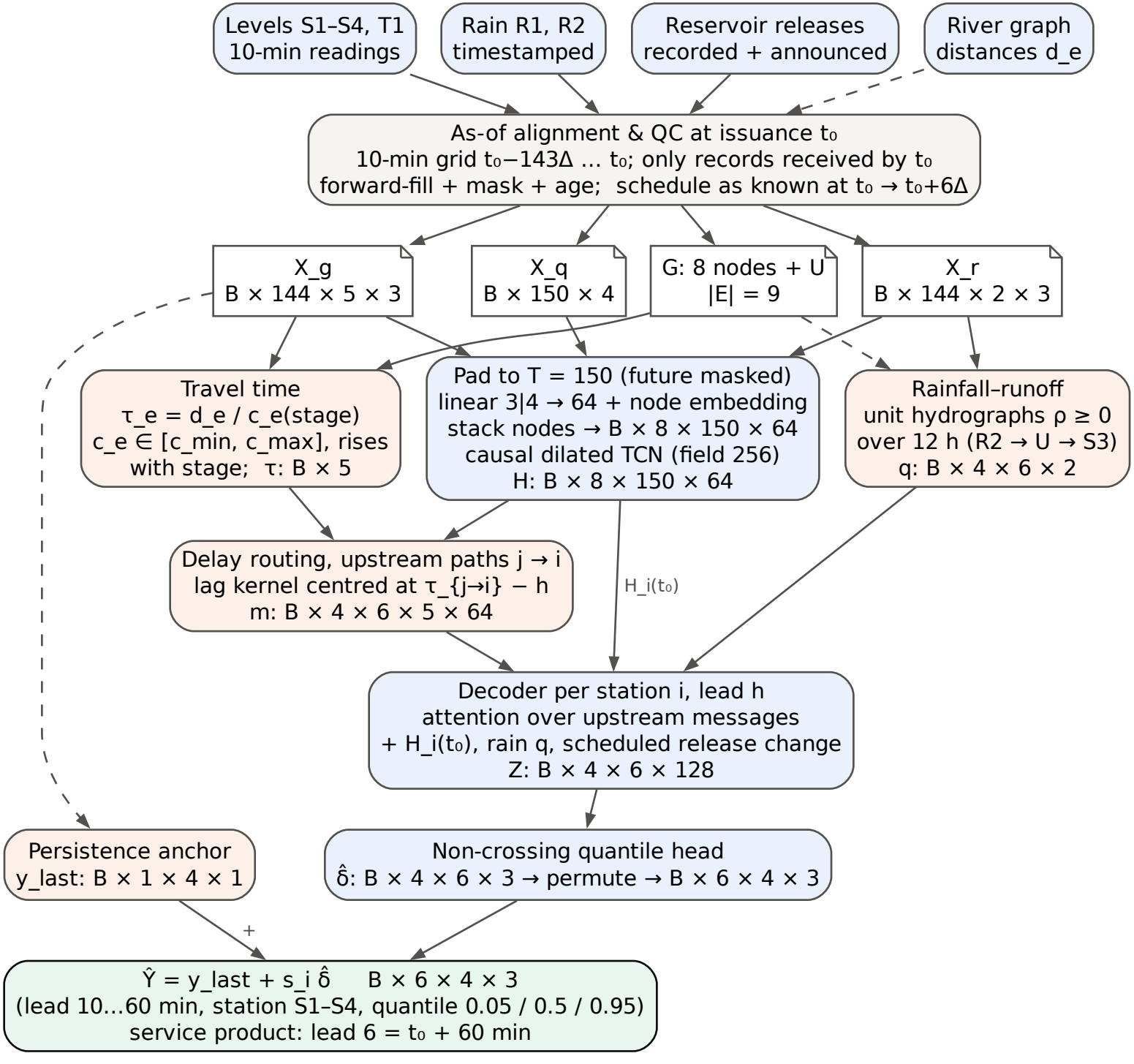
Levels S1–S4, T1
10-min readings
Rain R1, R2
timestamped
Reservoir releases
recorded + announced
River graph
distances d_e
As-of alignment & QC at issuance t₀
10-min grid t₀−143Δ … t₀; only records received by t₀
forward-fill + mask + age; schedule as known at t₀ → t₀+6Δ
X_g
B × 144 × 5 × 3
X_q
B × 150 × 4
G: 8 nodes + U
|E| = 9
X_r
B × 144 × 2 × 3
Travel time
τ_e = d_e / c_e(stage)
c_e ∈ [c_min, c_max], rises
with stage; τ: B × 5
Pad to T = 150 (future masked)
linear 3|4 → 64 + node embedding
stack nodes → B × 8 × 150 × 64
causal dilated TCN (field 256)
H: B × 8 × 150 × 64
Rainfall–runoff
unit hydrographs ρ ≥ 0
over 12 h (R2 → U → S3)
q: B × 4 × 6 × 2
Delay routing, upstream paths j → i
lag kernel centred at τ_{j→i} − h
m: B × 4 × 6 × 5 × 64
H_i(t₀)
Decoder per station i, lead h
attention over upstream messages
+ H_i(t₀), rain q, scheduled release change
Z: B × 4 × 6 × 128
Persistence anchor
y_last: B × 1 × 4 × 1
Non-crossing quantile head
δ̂: B × 4 × 6 × 3 → permute → B × 6 × 4 × 3
+
Ŷ = y_last + s_i δ̂ B × 6 × 4 × 3
(lead 10…60 min, station S1–S4, quantile 0.05 / 0.5 / 0.95)
service product: lead 6 = t₀ + 60 min